NUMBERS SYSTEMS

$$\begin{tikzcd} B \arrow[r, bend left, "\overset{t}{\text{Base}}"'] & S \end{tikzcd}$$

Element set $\quad$ symbol set

$a_i \in S$

$b \in B \qquad a_i \in \{0, 1, 2, \dots t-1\}$.

*Fixed point Number system

floating " " "

1) $b = s\,(a_n t^n + \cdots a_1 t + a_0 + a_{-1} t^{-1} + \cdots a_{-l} t^{-l}$

2) $b = s \sum_{i=-l}^{n} a_i t^i$

3) $b = s\, a_n a_{n-1} \cdots a_1 a_0 , a_{-1} \cdots a_{-l}$

$t=2$ Binary Number System $a_i \in \{0, 1\}$

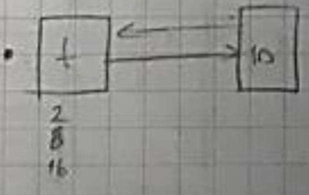
$t=8$ Octal " " $a_i \in \{0,1,2,3,4,5,6,7\}$

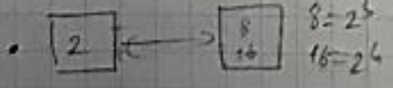
$t=10$ Decimal $a_i \in \{0, 1, 2 \dots 8, 9\}$

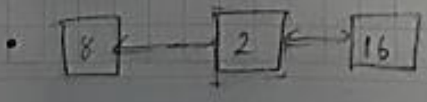
$t=16$ Hexadec $a_i \in \{0,1,2,3,4,5,6,7,8,9,$ $A, B, C, D, E, F\}$

$$b_t = \underset{\text{sign}}{s}\; \underbrace{\overset{\text{MSB}}{a_n} . a_{n-1} \cdots\cdots \overset{\text{LSB}}{a_0}}_{\text{integer}} , \underbrace{\underset{\text{MSB}}{a_{-1}} \cdots\cdots \underset{\text{LSB}}{a_{-l}}}_{\text{Fractional}}$$

BASE CONVERSION

- 

- 

- 

• t → 10

$b = (11010,10110)_2 \rightarrow$

$1 \cdot 2^4 + 1 \cdot 2^3 + 1 \cdot 2^1 + 1 \cdot 2^{-4}\, 1 \cdot 2^{-3} + 1 \cdot 2^{-4}$

$b = (26,6875)_{10}$

• 10 → t (2,8,16)

$b = N, M$

$N \Downarrow$ LSB ↓ MSB

$M \Downarrow$ MSB ↓ LSB

örn/ $(353)_{10} = (\;\ldots\;)_2$

$\overset{t}{2}$ 353	a_i	
176	1	a_0 LSB
88	0	
44	0	
22	0	
11	0	
5	1	
2	1	
1	0	
0	1	a_8 MSB

$$\begin{array}{r|l} A & t \\ \hline Q & B \end{array}$$

$(353)_{10} = (101100001)_2$

$A = B \cdot t + Q$

Fractional Conversion

$r^{-p} = t^{-q}$

$q = p \cdot \frac{\ln r}{\ln t} \Rightarrow q = p\,\frac{\ln 10}{\ln 2} \cong 3{,}32p$

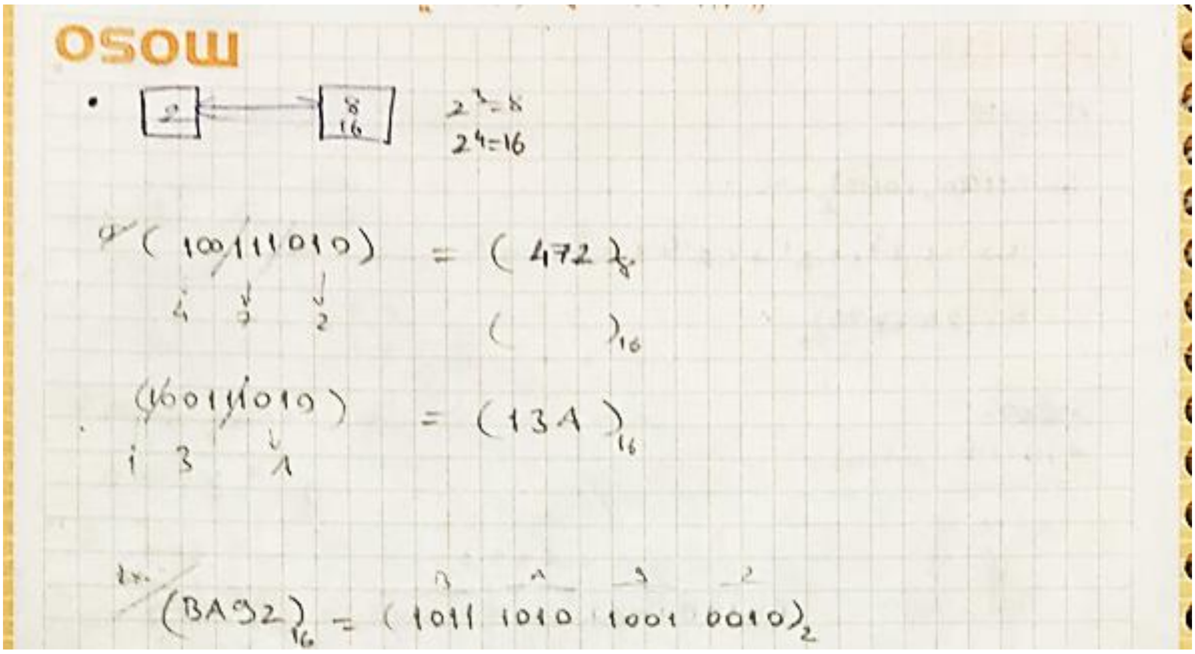
$M = (0{,}65625)_{10} = (\;\ldots\;)_2$

$q = 3{,}32 \cdot 5 = 16{,}6 \cong \boxed{17}$

eğer çıkmazsa
17 ye gidilir

	a_i	P	t
MSB	–	0,65625	2
	1	0,31250	
	0	0,6250	
	1	0,250	
	0	0,50	
LSB	1	0,0000	

$(0{,}10101)_2$



- $\boxed{2} \longleftrightarrow \boxed{\begin{matrix}8\\16\end{matrix}}$ $2^3=8$ $2^4=16$

$8\ (100/111/010) = (472)_8$

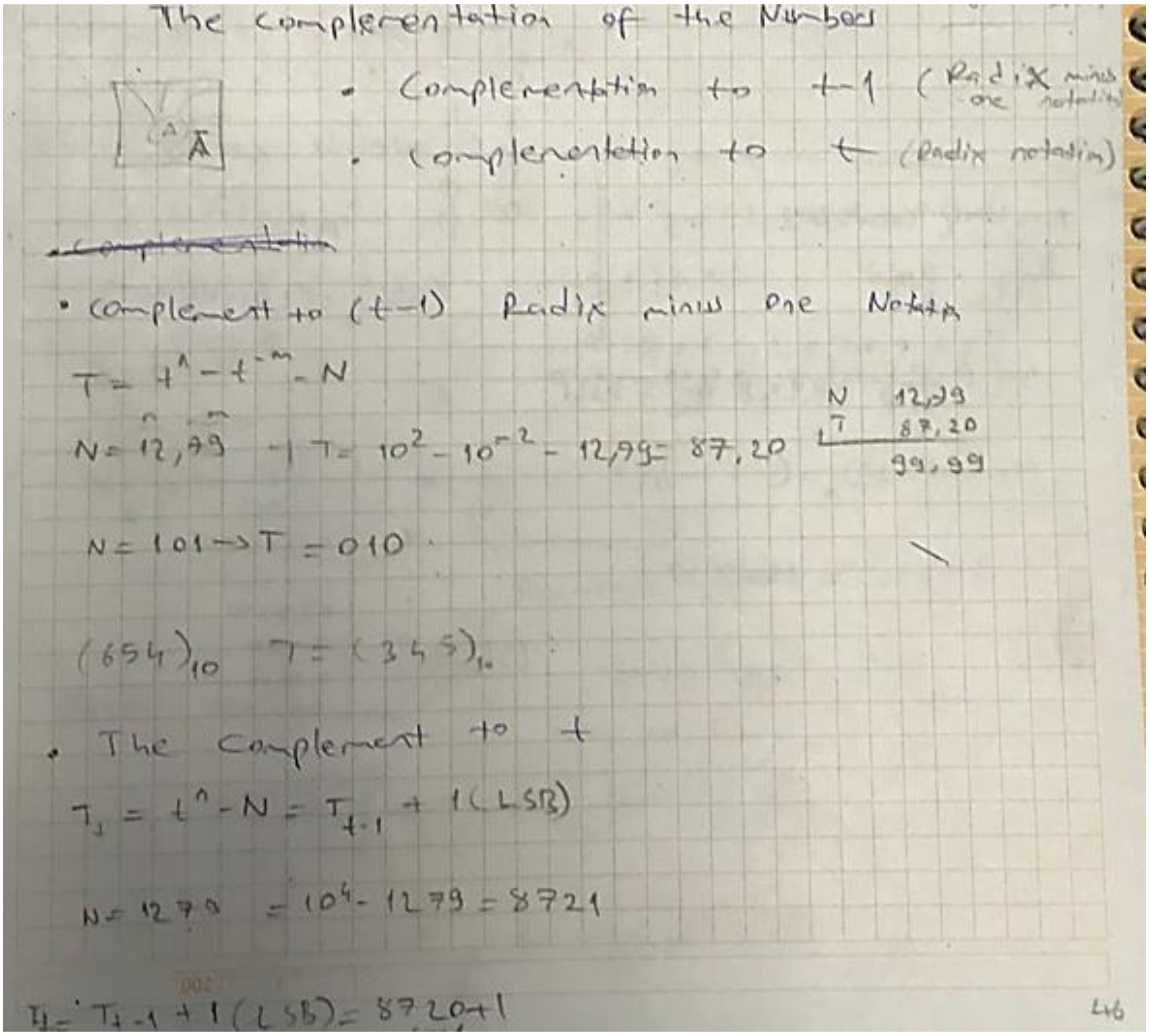
4 7 2

$(\quad)_{16}$

$(1/0011/1010) = (13A)_{16}$

1 3 A

Ex. $(BA92)_{16} = (1011\ 1010\ 1001\ 0010)_2$



The complementation of the Numbers

- Complementation to $t-1$ (Radix minus one notation)
- Complementation to t (Radix notation)

~~Complementation~~

- Complement to $(t-1)$ Radix minus one Notation

$T = t^n - t^{-m} - N$

$N = \overset{n}{12},\overset{m}{79} \rightarrow T = 10^2 - 10^{-2} - 12,79 = 87,20$

$\begin{array}{lr} N & 12,79 \\ +T & 87,20 \\ \hline & 99,99 \end{array}$

$N = 101 \rightarrow T = 010$

$(654)_{10}$ $T = (345)_{10}$

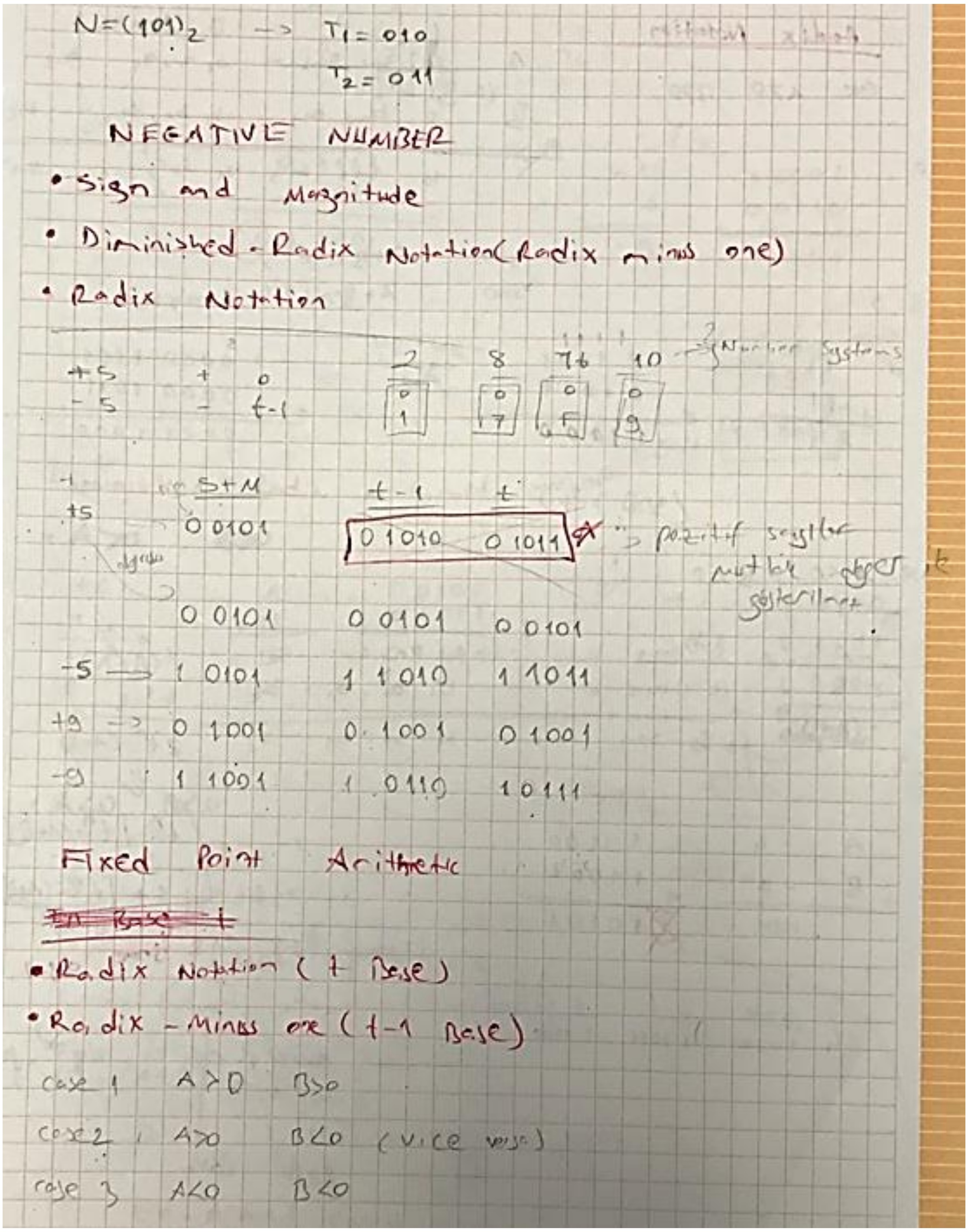
- The Complement to t

$T_t = t^n - N = T_{t-1} + 1 (LSB)$

$N = 1279 = 10^4 - 1279 = 8721$

$T_t = T_{t-1} + 1 (LSB) = 8720+1$

46



$N=(101)_2 \rightarrow T_1 = 010$

$T_2 = 011$

NEGATIVE NUMBER

- Sign and Magnitude
- Diminished - Radix Notation (Radix minus one)
- Radix Notation

			2	8	16	10
+5	+	0	0	0	0	0
−5	−	t-1	1	7	F	9

→ Number Systems

	S+M	t-1	t
+5	0 0101	0 1010	0 1011
	0 0101	0 0101	0 0101
−5 →	1 0101	1 1010	1 1011
+9 →	0 1001	0 1001	0 1001
−9	1 1001	1 0110	1 0111

~~0 1010 0 1011~~ → pozitif sayılar mutlak değer ile gösterilmez.

Fixed Point Arithmetic

~~En Base t~~

- Radix Notation (t Base)
- Radix - Minus one (t-1 Base)

case 1 A > 0 B > 0

case 2 A > 0 B < 0 (vice versa)

case 3 A < 0 B < 0

Radix Notation

case A>0 B>0

$$\begin{array}{rl} A & 0\ a_{n-2}\ a_{n-3} \cdots a_1\ a_0,\ a_{-1} \cdots a_{-t} \\ + \ B & 0\ b_{n-2}\ b_{n-3} \cdots b_1\ b_0,\ b_{-1} \cdots b_{-t} \\ \hline & s_s\ s_{n-2}\ s_{n-3}\ s_1\ s_0,\ s_{-1} \cdots s_{-t} \end{array}$$

$s_s = 0$ $A+B < t^{n-1}$

$s \neq 0$ $A+B > t^{n-1}$ (overflow)

Ör

$$\begin{array}{r} 13 \\ +\ 11 \\ \hline 24 \end{array} \qquad \begin{array}{r} 1111 \\ 01101 \\ +\ 01011 \\ \hline \boxed{1}1000 \end{array}$$

Overflow

8 bit yer ayırmış olsa →

$$\begin{array}{r} \overset{s}{0}000\ 1101 \\ +\ 0000\ 1011 \\ \hline 0001\ 1000 \end{array}$$

↑ (x)

Case A<0 , B<0

$$\begin{array}{rll} -A & (t-1)\ a_{n-2} \cdots a_1\ a_0,\ a_{-1} \cdots a_{-t} & t^n - A \\ +\ -B & (t-1)\ b_{n-2} \cdots b_1\ b_0,\ b_{-1} \cdots b_{-t} & t^n - B \\ \hline -(A+B) & C_s\ S_s\ s_{n-2} \cdots s_1\ s_0,\ s_{-1} \cdots s_{-t} & 2t^n - (A+B) \end{array}$$

carry

$$\Downarrow$$

$$t^n\ [t^n-(A+B)]$$

Carry Bit → $1 + \left[\frac{t^n-(A+B)}{t^n}\right]$

ignore digit

$$\begin{array}{rrr} A & -4 & 11100 \\ +\ B & -7 & 11001 \\ \hline & -11 & \cancel{1}10101 \end{array}$$

⇒ take 2's complement ⇒ $\overset{\text{Sign}}{\boxed{11011}}$

-4 $\quad \frac{s+M}{10100} \quad \frac{t-1}{11011} \quad \frac{t}{11100}$

işaret ve mutlak değer göster

$$\underset{t}{10101} \rightarrow \underset{\cancel{-1}}{11010} \rightarrow \frac{(S+M)}{11011}$$

$$\frac{10100}{t-1}$$

Şunası [illegible] için

b-

$A > 0$, $B < 0$

$$\begin{array}{rr} A & A \\ + \; -B & t^n - B \\ \hline & t^n - (B-A) \end{array}$$

ö/

$$\begin{array}{rr} 15 & 0\ 1111 \\ -6 & 1\ 1010 \\ \hline 9 & \cancel{1}\ 0\ 1001 \end{array}$$

ö/

$$\begin{array}{rr} -15 & 1\ 0001 \\ + \; 6 & + \; 0\ 0110 \\ \hline -9 & 1\ 0111 \end{array}$$

↓ t-1e 2's com

↓ t-1

1000

↓ +1

S+M $\boxed{1\ 1001}$

Diminished - Radix Arithmetic (t-1 Base)

- $A > 0$, $B > 0$

$$\begin{array}{rrr} +A & 4 & 0\ 0100 \\ +B & + \; 11 & + \; 0\ 1011 \\ \hline +(A+B) & +15 & 0\ 1111 \end{array}$$

↑

0

1

ö/

$$\begin{array}{rr} +6 & 0\ 0110 \\ + \; +11 & + \; 0\ 1011 \\ \hline & 1\ 0001 \end{array}$$

↑ overflow

False

- $A < 0$ $B < 0$

$$\begin{array}{rr} -A & t^n - t^{-k} - A \\ + \; -B & + \; t^n - t^{-k} - B \\ \hline -(A+B) & 2t^n - 2t^{-k} - (A+B) \end{array}$$

$$t^n - t^{-k} + [t^n - t^{-k} - (A+B)]$$

⇓

$t^n - t^{-k} - (A+B)$

MSB LSB

End - Around Carry

Arithmetic Circuits

Half Adder

Full Adder

Full Substracter.

Multiplier

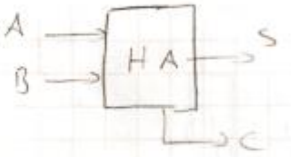
Divider

Comparator.

OSOM

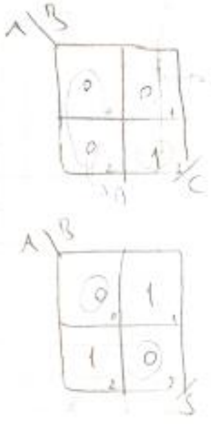
Half Adder

a) uygulayın NAND
b) '' NOR kapıları ile gerçekleştir



$$\begin{array}{r} A \\ + B \\ \hline CS \end{array}$$

Dec	A	B	C	S
0	0	0	0	0
1	0	1	0	1
2	1	0	0	1
3	1	1	1	0

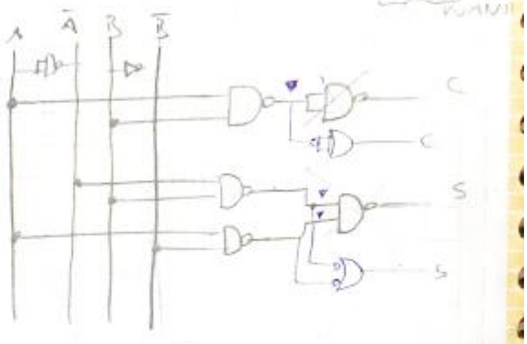

A \ B
0
0
0
1
C
A \ B
0
1
1
0
S

a) $C = AB$

$S = \bar{A}B + A\bar{B}$

$C = \overline{\overline{AB}}$

$S = \overline{\overline{\bar{A}B + A\bar{B}}} = \overline{\underbrace{\overline{\bar{A}B} \cdot \overline{A\bar{B}}}_{NAND}}$



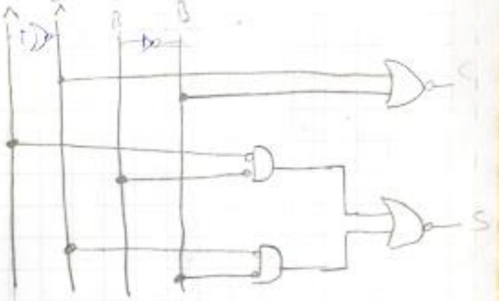
b). $C = A \cdot B$

$S = (A+B) \cdot (\bar{A}+\bar{B})$

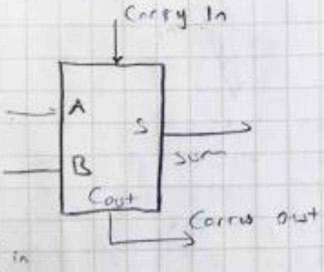
$C = \overline{\overline{A \cdot B}} = \overline{\bar{A} + \bar{B}}$

$S = \overline{\overline{(A+B) \cdot (\bar{A}+\bar{B})}} =$

$= \overline{\overline{A+B} + \overline{\bar{A}+\bar{B}}}$

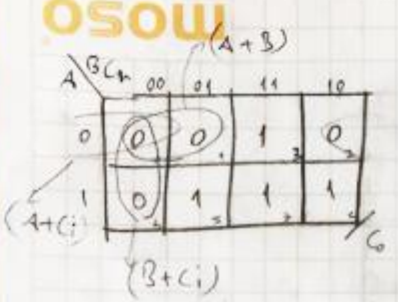


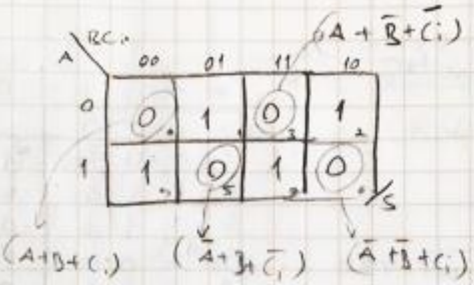
Full Adder


Carry In
A
B
S
sum
Cout
Carry out

$$\begin{array}{r} C_{in} \\ A \\ B \\ \hline C_{out}\,S \end{array}$$

Dec	A	B	Cin	Cout	S
0	0	0	0	0	0
1	0	0	1	0	1
2	0	1	0	0	1
3	0	1	1	1	0
4	1	0	0	0	1
5	1	0	1	1	0
6	1	1	0	1	0
7	1	1	1	1	1


(A+B)
A
BC_{in}
00
01
11
10
0
1
0
0
1
0
0
1
1
1
$(A+C_i)$
$(B+C_i)$
C_o

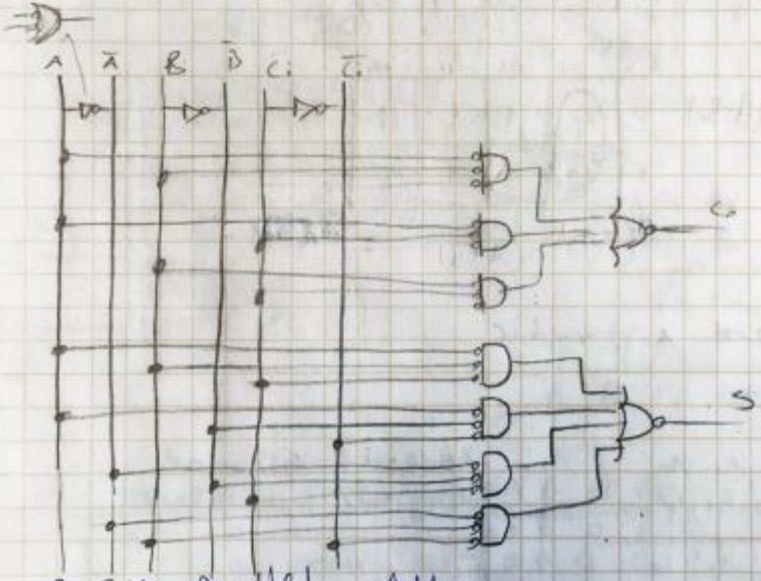

$(A+\bar{B}+\bar{C_i})$
A
BC_i
00
01
11
10
0
1
0
1
0
1
1
0
1
0
S
$(A+B+C_i)$
$(\bar{A}+B+\bar{C_i})$
$(\bar{A}+\bar{B}+C_i)$

$$C_o = (A+B)\cdot(A+C_i)\,(B+C_i)$$

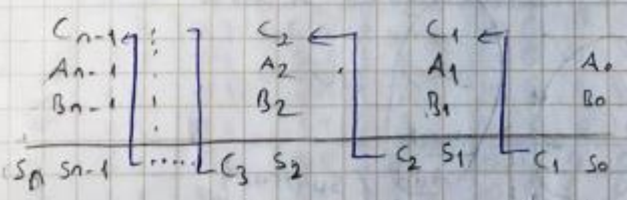
$$= \overline{\overline{(A+B)} + \overline{(A+C_i)} + \overline{(B+C_i)}}$$

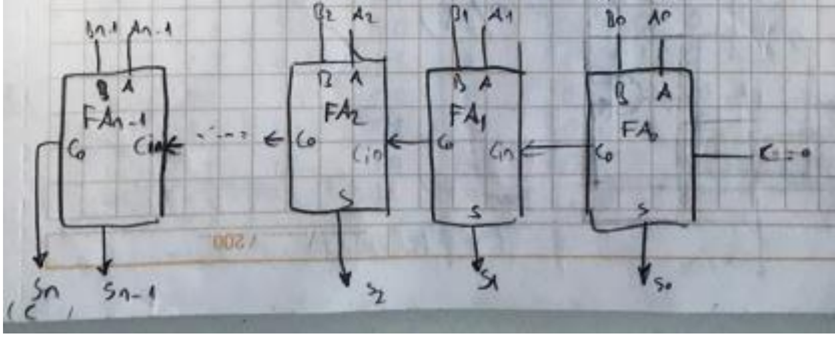
$$S = (A+B+C_i)\cdot(A+\bar{B}+\bar{C_i})\cdot(\bar{A}+\bar{B}+C_i)(\bar{A}+B+\bar{C_i})$$

$$\overline{\overline{(A+B+C_i)} + \overline{(A+\bar{B}+\bar{C_i})} + \overline{(\bar{A}+\bar{B}+C_i)} + \overline{(\bar{A}+B+\bar{C_i})}}$$

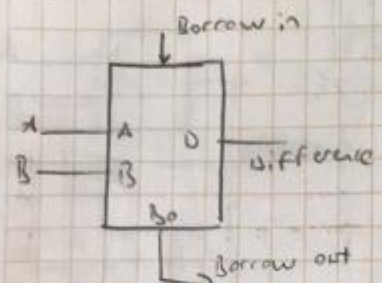

A
$\bar{A}$
B
$\bar{B}$
C_i
$\bar{C_i}$
C_o
S

n Bit parallel Adder


C_{n-1}
A_{n-1}
B_{n-1}
C_2
A_2
B_2
C_1
A_1
B_1
A_0
B_0
S_n
S_{n-1}
C_3
S_2
C_2
S_1
C_1
S_0


B_{n-1}
A_{n-1}
B
A
FA_{n-1}
C_o
C_{in}
B_2
A_2
B
A
FA_2
C_o
C_{in}
S
B_1
A_1
B
A
FA_1
C_o
C_{in}
S
B_0
A_0
B
A
FA_0
C_o
S
$C_i=0$
S_n
S_{n-1}
S_2
S_1
S_0

Full Subtractor


Borrow in
A
B
A
B
D
Difference
Bo
Borrow out

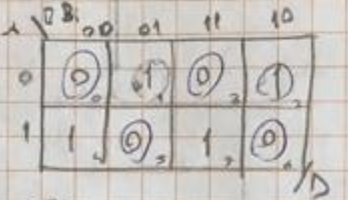
Dec	A	B	Bin	D	Bo
0	0	0	0	0	0
1	0	0	1	1	1
2	0	1	0	1	1
3	0	1	1	0	1
4	1	0	0	1	0
5	1	0	1	0	0
6	1	1	0	0	0
7	1	1	1	1	1

$-B_{in}$

A

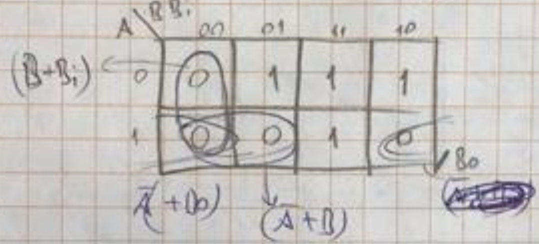
$-B$

$B_o \; D$



A \ B Bi	00	01	11	10
0	0	1	0	1
1	1	0	1	0

D



A \ B Bi	00	01	11	10
0	0	1	1	1
1	0	0	1	0

(B+Bi)
Bo
Ā(+Bo)
(A+B)

Sign complement Arithmetic

Adder / Subtractor

$A_{n-1} \ldots\ldots A_2\ A_1\ A_0$

$B_{n-1} \ldots\ldots B_2\ B_1\ B_0$

$C_n'\ S_{n-1} \ldots\ldots S_2'\ S_1'\ S_0'$

→ Subtract 1's complement plus C_n'

$S_n\ S_{n-1} \ldots\ldots S_2\ S_1\ S_0$

gnored
subtract
's complement
Add

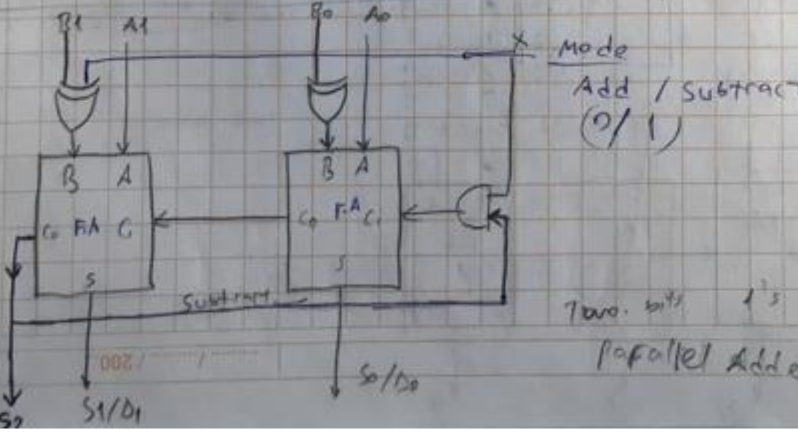
Addend / Minuend

Augend / subtrahend

first sum

Carryend Arand and Add

Sum / Difference

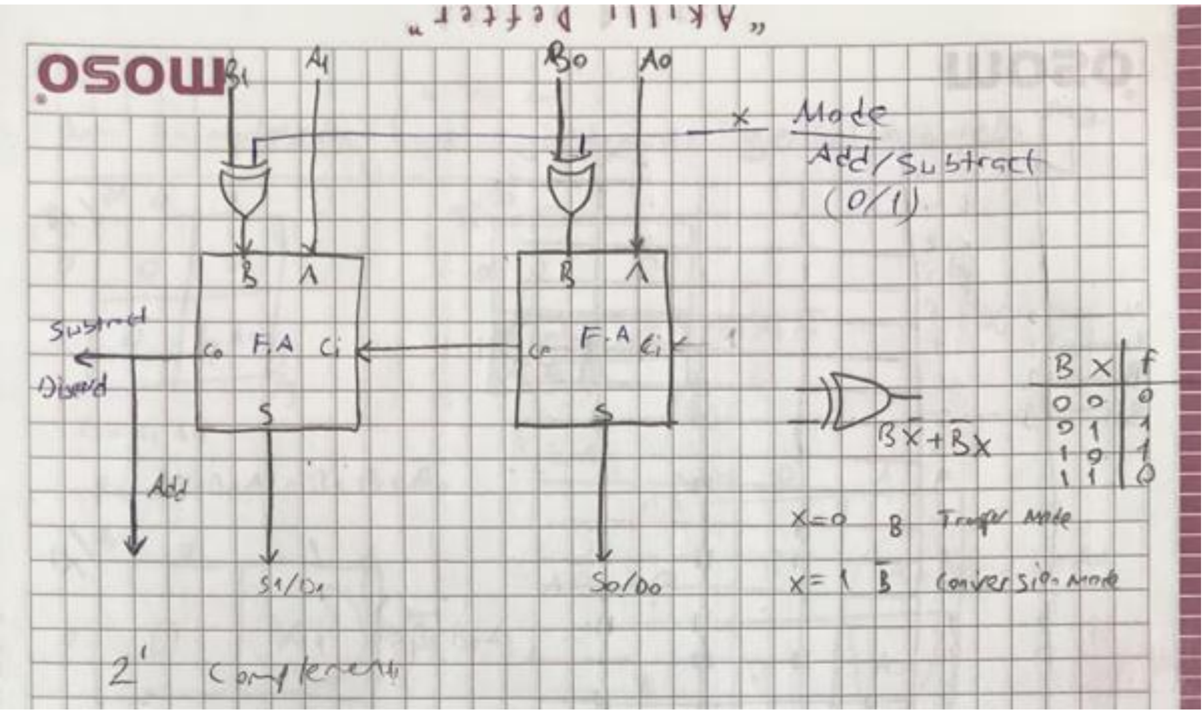

B1
A1
B0
A0
x
B
A
C0
F.A
C
S
B
A
C0
F.A
C1
S
Subtract
Add
S2
S1/D1
S0/D0

Mode

Add / Subtract (0/1)

Two. bits 1's complement

Parallel Adder / Subtractor.


B1
A1
B0
A0
X
B
A
F.A
Co
Ci
S
Subtract
Discard
Add
S1/O1
S0/O0
1

Mode
Add/Substract
(0/1)

$B\bar{X}+\bar{B}X$

B	X	f
0	0	0
0	1	1
1	0	1
1	1	0

X=0 B Transfer Mode

X=1 $\bar{B}$ Conversion Mode

2' Complement